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The Effect of Pulse Duration on Properties of Boron Nanoparticles Produced by Laser Fragmentation of Micropowders in Liquids

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Abstract—Optimization of laser synthesis regimes producing colloidal solutions of boron nanoparticles for their application as a promising boron-containing agent of boron neutron capture therapy (BNCT) of oncological diseases is an up to date trend in BNCT advancement. In this study, colloidal solutions of boron nanoparticles (BNPs) were synthesized by laser fragmentation in organic solvents using laser pulses of nanosecond and femtosecond duration range. The powder of the stable ^{10}B isotope of submicron fractions with arbitrary surface morphology served as starting material in both cases. The dependence of the properties of produced nanomaterials on the duration of laser pulses was compared directly.

Keywords: boron nanoparticles, laser ablation in liquid, laser synthesis, pulsed fragmentation in liquid, binary therapy

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1. INTRODUCTION

The use of nanoparticles (NP) in targeted radionuclide therapy procedures and in binary radiotherapy methods is a promising area that can improve the efficiency and reduce side effects and the impact of ionizing radiation on healthy organs [1–3]. Boron (B)-based nanoparticles are widely used as sensitizers in binary hadron therapies including therapy with proton beams [4–6] and using thermal and epithermal neutron beams [7–9]. Compared to molecular boron-containing drugs for boron-neutron-capture therapy (BNCT) designed to play this role [10], pure boron NPs carry the maximum number of atoms that effectively interact with the radiation thus used.

Methods of laser ablation synthesis in liquid can ensure record surface cleanliness of produced nanomaterials because they do not require chemical precursors that contaminate the colloidal solution [11–13]. Laser fragmentation of elemental ^{10}B micropowders is one of these methods [14]. The interest in this method is due to easy availability of micropowders of the required isotopic composition. At the same time, the physical-chemical properties of the produced nanoparticles can vary significantly depending on the range of laser pulse durations and irradiation conditions leaving a wide field for research [15]. The question of choosing the optimal synthesis technology remains unresolved both in terms of the properties of resulting materials and productivity of the method and in terms of the cost of the equipment to be used. This paper compares approaches using two high repetition rate setups of short, nanosecond (ns) and ultrashort, femtosecond (fs) laser pulses. The task was to produce NP fractions of interest for biomedical applications using the same starting micropowder of the stable ^{10}B isotope and to compare physical-chemical properties of produced nanoparticles when using laser pulses of different durations.

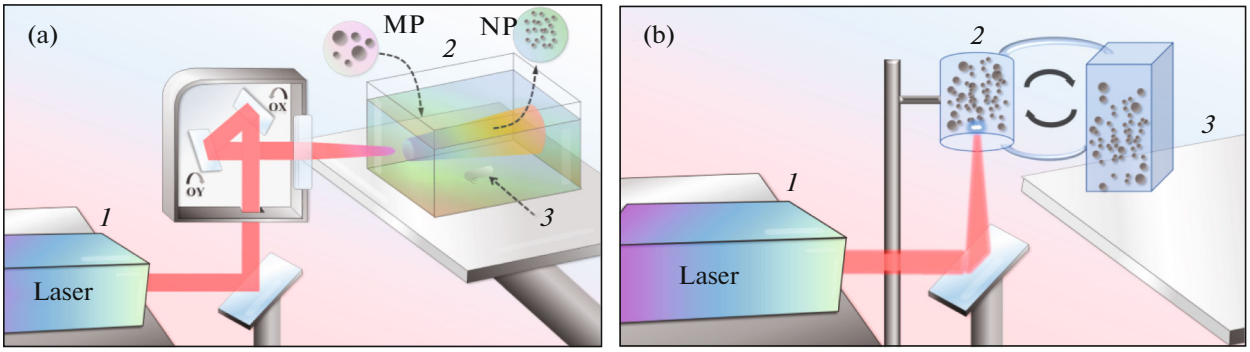

Fig 1. The experimental setup for laser fragmentation of ^{10}B micropowders: (*1*) fs laser source, (*2*) suspension of ^{10}B microparticles in acetonitrile, (*3*) magnetic stirrer for mixing liquids (a); (*1*) ns laser source with F-theta lens, (*2*) ^{10}B MP suspension in isopropyl alcohol, (*3*) peristaltic pump with flowthrough cuvette (b).

2. MATERIALS AND METHODS

The stable ^{10}B isotope powder with 85% enrichment was used as a starting material. For the laser fragmentation procedure, the powder was dispersed in liquid at a concentration of 100 μg/mL. Organic solvents were used in both cases thus preventing oxidation of the particles while ensuring high purity and sterility of the produced nanomaterial. The synthesis was carried out with high repetition rate of laser pulses, femtosecond or nanosecond, under vigorous stirring of the solution or its pumping through the focusing region. The colloidal solution produced by laser fragmentation was centrifuged to obtain comparable size distributions. The features of nanomaterial synthesis using two durations of laser pulses are presented below.

2.1. Fragmentation with Femtosecond Pulses

Some boron nanoparticles were produced by the fs-laser fragmentation method in liquid, which we had developed in previous studies [16–19]. These nanoparticles are hereafter denoted fs-B. The Yb:KGW laser TETA-20 (1030 nm wavelength, 270 fs pulse duration, 75 μJ pulse energy, 100 kHz repetition rate, Avesta, Russia) was used as a source of laser radiation for their synthesis. The experimental setup is shown in Fig. 1a. Laser radiation was directed by a system of mirrors to a galvanoscanner (LScan-10, Ateko-TM, Moscow, Russia) and then focused by an F-Theta lens (F = 100 mm) through the side wall in a glass cuvette filled with 20 mL of liquid. A magnet was also placed inside the cuvette, and the solution cuvette itself was mounted on a magnetic stirrer that rotated the magnet at 500 rpm and mixed the suspension during laser treatment. The position of the laser beam was fixed 10 mm below the air-liquid interface, and the position of the beam waist was shifted inside the cuvette about 10 mm from the entrance wall and did not change throughout the laser treatment time. Acetonitrile was used as the liquid medium for laser fragmentation. It was performed at room temperature. The resulting colloidal solution was then completely withdrawn from the cuvette and centrifuged at 6000*g* for 1.5 min to separate the NPs by size. The supernatant obtained during this procedure was collected and used for further experiments.

2.2. Fragmentation with Nanosecond Pulses

The second part of boron nanoparticles was produced by ns laser fragmentation in isopropanol in a flowthrough cuvette. These nanoparticles are further denoted as ns-B. The experimental setup is shown in Fig. 1b. Nd:AG-laser (Bright Solutions SRL, Italy) was used as a radiation source with the following characteristics: 1064 nm wavelength, 10 ns pulse duration, 1 mJ energy in pulses, and 10 kHz pulse repetition rate. The working fluid volume was 70 mL. The choice of working fluid was justified by its low reactivity towards boron. Several preliminary experiments found that the laser fragmentation rate of powders of various materials was maximized when using a flowthrough cuvette under constant stirring of the colloid thus increasing the proportion of particles exposed to the laser effect. In order to produce ns-B in isopropanol, the micropowder suspension was exposed to a focused laser beam directed bottom-up through the glass window of the cuvette. The laser beam waist was inside the cuvette a few millimeters above the window. The calculated diameter of the laser beam in the waist was 50 μm. The laser light was moved by a galvano optical system at 500 mm/s and focused by an F-Theta lens (F = 90 mm). For additional selec-

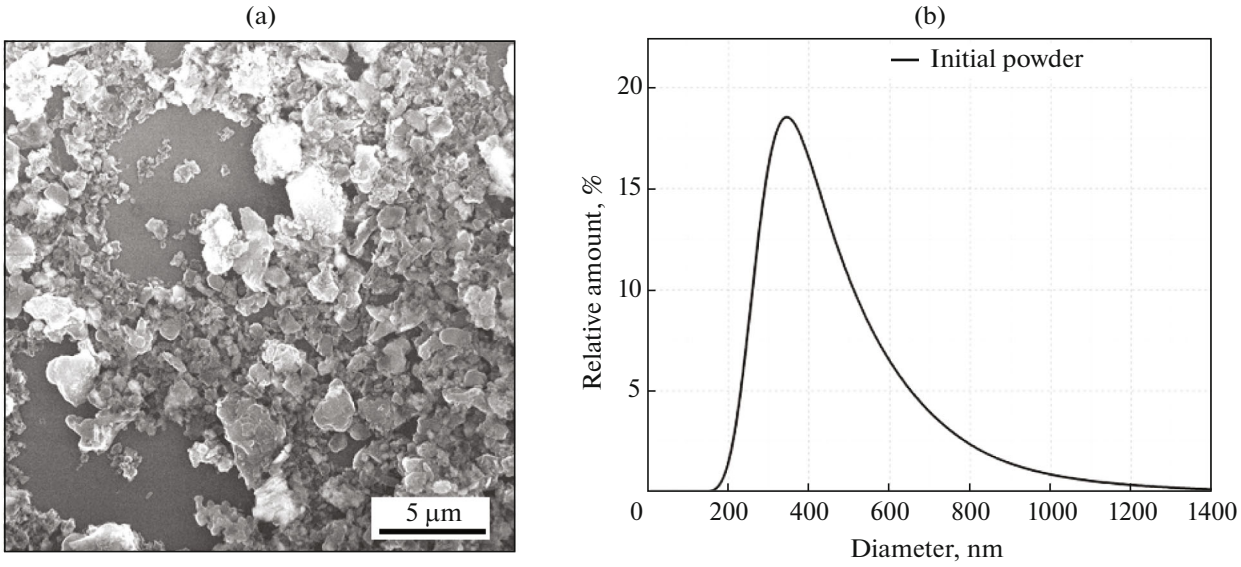

Fig. 2. SEM image (a) and hydrodynamic dimensions of boron powder prior to fragmentation (b).

tion by size of ns-B nanoparticles, they were centrifuged for 1, 5, 7, and 10 min at 3000*g*. All large precipitated NPs were removed from the experiment and the supernatant that did not precipitate was taken for further studies.

2.3. Nanoparticle Characterization Methods

The morphology and size of all particles were characterized using a scanning electron microscopy (SEM) system (MAIA 3, Tescan, Czech Republic) and transmission electron microscopy (TEM) (Carl Zeiss, Jena, Germany). Electron SEM and TEM images were obtained at accelerating voltages of 30 and 200 keV, respectively. Samples for measurements were prepared by applying 2 μL of colloid on a cleaned single-crystal silicon substrate (SEM) or 1 μL of colloid on a carbon grid (TEM) followed by drying at room temperature. Size distributions of boron nanoparticles were obtained by analyzing electron microscopy images in the ImageJ software environment highlighting manually individual nanoparticles. Each distribution was based on diameter measurements of at least 300 nanoparticles.

The hydrodynamic size of the particles was measured by the dynamic light scattering (DLS) method using a Zetasizer Nano ZS analyzer (Malvern Instruments, UK). The measurements were made in glass cuvettes with an optical path of 10 mm. Numerically weighted size distributions were used for the analysis.

3. RESULTS AND DISCUSSION

The morphology and particle size of starting powder were measured prior to the laser fragmentation process. The results of these measurements are shown in Fig. 2. As can be seen from the obtained data, the untreated powder particles were randomly shaped with their sizes ranging from a few hundred nanometers to a few micrometers (Fig. 2a). The measurements of the hydrodynamic size distribution of these particles presented in Fig. 2b showed that the distribution mode was around 340 nm (full width at half-height of 250 nm).

One can notice that the hydrodynamic size distribution does not contain particles larger than 1.5 microns, which are visible, however, in the SEM image. These differences in the data are due to the specifics of the DLS method. Measurements with this method take a few minutes, during which time large powder particles can settle at the bottom of the test tube and they are not included in the final distribution.

Then we checked the dependence of the dimensional properties of nanoparticles on fragmentation time separately for pulses of ns and fs durations. In both cases, a longer duration of the laser fragmentation process led to both the shorter mode and width of size distribution, but the characteristic times during which the changes occurred and the values of these changes differed (Fig. 3). In the case of fs laser exposure, the mode and full width at half-height decreased in 24 min from initial values (340 and 250 nm,

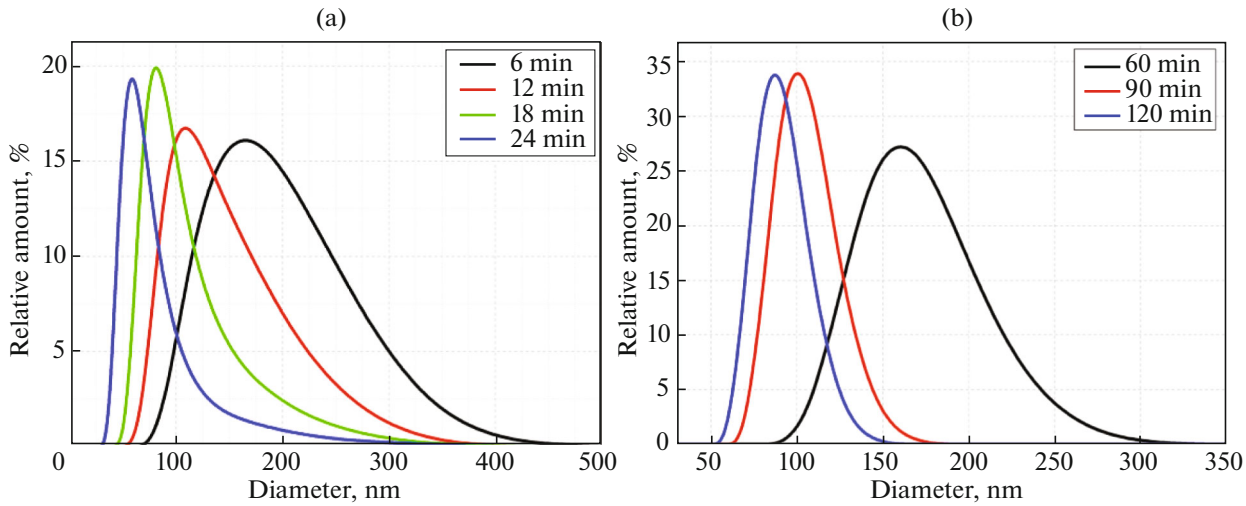

Fig. 3. Dynamics of change in the hydrodynamic size of nanoparticles depending on femtosecond (a) and nanosecond laser fragmentation time (b).

respectively) to 59 and 40 nm. When ns laser pulses were used, the size of nanoparticles dropped to 80 nm in 120 min.

Further reduction in the size of ns-B nanoparticles to values achievable by fs fragmentation required an additional post-treatment of the nanoparticles by centrifuging. We studied the dependence of the size distribution parameters of ns-B nanoparticles on the duration of centrifugation. It was found by analyzing SEM images and size distributions built from them that centrifugation of ns-B nanoparticles for 1, 5, 7, and 10 min at 3000*g* leads to a consistent decrease in their average size from 61 to 38 nm. And in the process the average size of ns-B nanoparticles almost equaled that of fs-B nanoparticles (53 and 59 nm, respectively) when centrifuged for 5 min. The focus on achieving just this average size was driven by plans of biological applications of boron nanoparticles. The point is that the size of nanoparticles strongly affects the efficiency of their cellular internalization and the duration of circulation in the bloodstream [20]. The optimum size of nanoparticles for in vivo applications is generally considered to be around 50 nm [21]. This size ensures efficient cellular internalization of nanoparticles and allows to avoid their fast elimination from the body through the renal glomerular filtration system. In addition, the efficiency of their removal from the bloodstream by the macrophage system increases with the size of nanoparticles [22], which makes it difficult to use larger NPs.

Figure 4 shows data on final size distribution of nanoparticles and their SEM images. As can be seen from these findings, the nanoparticles had a spherical shape. In the case of fs pulses, the mode of the size distribution of final nanoparticles was 48 nm and it was 53 nm when ns pulses were used.

It should be noted that under fs fragmentation the size distribution maxima obtained by DLS and SEM actually coincide while under ns irradiation the maxima differ being 80 nm for DLS and 53 nm for SEM. The probable cause of the observed effect is the formation of a carbon shell around ns-B nanoparticles (Fig. 5), which is the product of isopropanol decomposition when the working fluid is exposed to laser radiation. This shell changes the optical properties of nanoparticles, which significantly affect the dimensions measured by dynamic light scattering.

Also, in the case of ns fragmentation the resulting product has to be “washed” by centrifuging due to the decomposition of isopropyl alcohol. This is due to laser breakdown around the nanoparticles. IR laser radiation is absorbed during exposure of organic liquids, such as alcohols, by impurities contained in the liquid or, for example, those getting into the cuvette from ambient air. The absorbed radiation ionizes nanoparticles, which results in the appearance of plasma plumes around them and hence in plasma radiation. When organic liquids are used, the heating and exposure to plasma radiation leads to the decay of alcohol molecules. Carbon nanostructures appear in the liquid during the decay process, which can interact with boron nanoparticles to form a carbon shell. The main factor affecting the appearance of this shell is the plasma lifetime, which depends above all on the duration of laser pulses. This is the reason why the carbon shell is absent in fs-B nanoparticles.

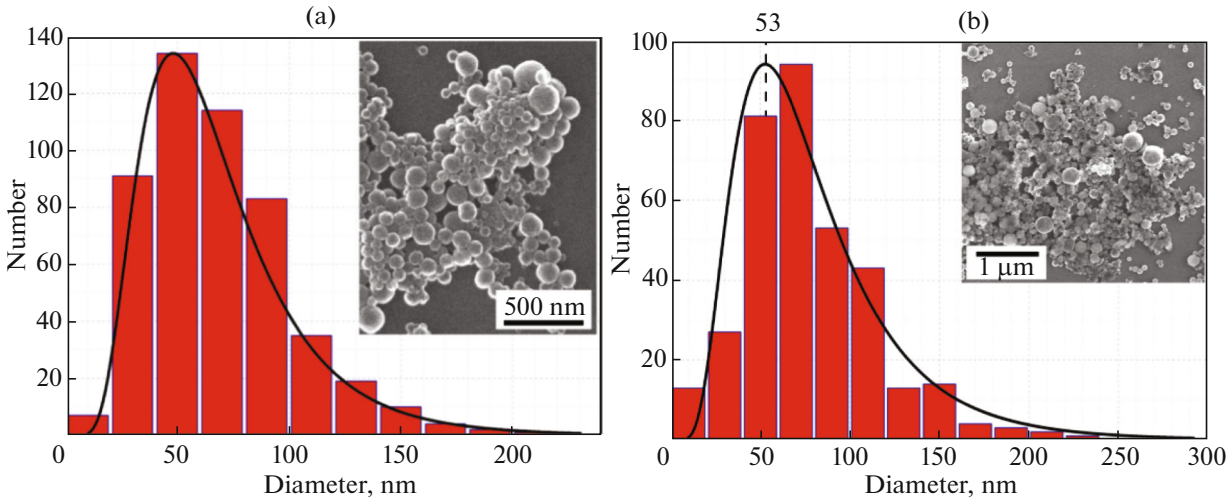

Fig. 4. Size distributions and SEM images of nanoparticles after laser fragmentation for fs-B (a) and ns-B (b).

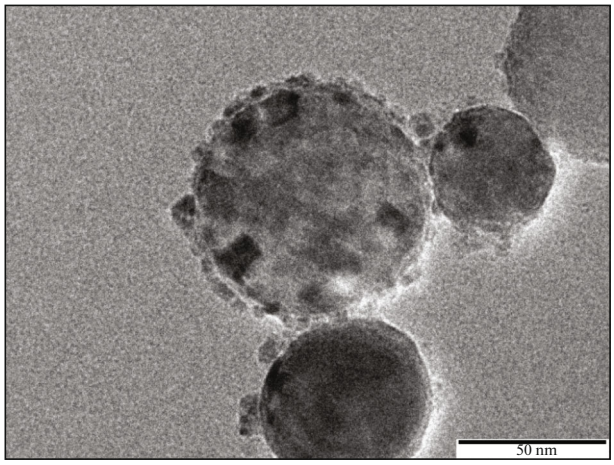

Fig. 5. TEM image of ns-B nanoparticles.

4. CONCLUSIONS

The dimensional and morphological properties of ^{10}B nanoparticles synthesized by laser fragmentation in liquid were investigated in this study. A comparative analysis was made of nanoparticle properties resulted from fragmentation by fs and ns pulses, which showed that laser fs fragmentation of boron powder was faster than ns fragmentation. Differences in the chemical compositions of the surface were noted due to the duration of laser exposure, namely, the nc-B nanoparticles had carbon shells while they were absent in the fc-B nanoparticles. The question remains open as to the productivity of synthesizing colloidal solutions of B nanoparticles of a given size to provide the necessary volumes of material for application. Additional research will be conducted to optimize laser modes to ensure the productivity.

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CONFLICT OF INTEREST

The authors declare that they have no conflicts of interest.

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